SOLUSI LATIHAN SLIDE INFORMATION REPRESENTATION DDAK 2022/2023

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PERHATIAN
- Jangan baca solusi sebelum mengerjakan soal pada slide
- Jika ada hal yang kurang jelas dari jawaban harap email ke: **herik@cs.ui.ac.id**

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2-5. Convert each of the following decimal numbers to binary (base two) with at most eight digits in the fractional part, rounded to eight places.
a. 2000 b. 0.875 c. 0.07 d. 12.345

Solusi:

(a) $(2000)_{10} = (11111010000)_2$
(b) $(0.875)_{10} = (0.111)_2$
(c) $(0.07)_{10} = (0.00010001)_2$
(d) $(12.345)_{10} = (1100.01011000)_2$

2-6. Convert each of the decimal numbers in Question 2-5 above to septimal (base seven) with at most six digits in the fractional part, rounded to six places.

Solusi:

(a) $(2000)_{10} = (5555)_7$
(b) $(0.875)_{10} = (0.606061)_7$
(c) $(0.07)_{10} = (0.033003)_7$
(d) $(12.345)_{10} = (15.226223)_7$

2-7. Convert each of the decimal numbers in Question 2-5 above to octal (base eight) with at most four digits in the fractional part, rounded to four places.

Solusi:

(a) $(2000)_{10} = (3720)_8$
(b) $(0.875)_{10} = (0.7)_8$
(c) $(0.07)_{10} = (0.0437)_8$
(d) $(12.345)_{10} = (14.2605)_8$

2-8. Convert each of the decimal numbers in Question 2-5 above to hexadecimal (base sixteen) with at most two digits in the fractional part, rounded to two places.

Solusi:

(a) $(2000)_{10} = (7D0)_{16}$
(b) $(0.875)_{10} = (0.E)_{16}$
(c) $(0.07)_{10} = (0.12)_{16}$
(d) $(12.345)_{10} = (C.58)_{16}$

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2-9. Which of the following octal values is equivalent to the binary number $(110001)_2$?
a. $(15)_8$ b. $(31)_8$ c. $(33)_8$ d. $(49)_8$ e. $(61)_8$

Solusi:

e.$(61)_8$

2-10. Convert the binary number $(1001101)_2$ to
a. quaternary b. octal c. decimal d. hexadecimal

Solusi:

(a) $(1031)_4$ (b) $(115)_8$ (c) $(77)_{10}$ (d) $(4D)_{16}$

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What is $(1011)_2 \times (101)_2$?
a. $(10000)_2$ b. $(110111)_2$ c. $(111111)_2$ d. $(111011)_2$ e. $(101101)_2$

Solusi:

(b) (110111)2

Perform the following operations on binary numbers.
a. $(10111110)_2 + (10001101)_2$
b. $(11010010)_2 - (01101101)_2$
c. $(11100101)_2 - (00101110)_2$

Solusi:

(a) $(101001011)_2$ (b) $(01100101)_2$ (c) $(10110111)_2$

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2-13. In a 6-bit 2's complement binary number system, what is the decimal value represented by $(100100)_{2s}$?
a. –4 b. 36 c. –36 d. –27 e. –28

Solusi:

e. -28

2-14. In a 6-bit 1's complement binary number system, what is the decimal value represented by $(010100)_{1s}$?

a. –11 b. 43 c. –43 d. 20 e. –20

Solusi:

d. 20

2-15. What is the range of values that can be represented in a 5-bit 2's complement binary system?

a. 0 to 31 b. –8 to 7 c. –8 to 8 d. –15 to 15 e. –16 to 15

Solusi:

e. -16 to 15

2-16. In a 4-bit 2's complement scheme, what is the result of this operation: $(1011)_{2s} + (1001)_{2s}$?

a. 4 b. 5 c. 20 d. –12 e. overflow

Solusi:

e. overflow

2-17. Assuming a 6-bit 2's complement system, perform the following subtraction operations by converting it into addition operations:

a. $(011010)_{2s} - (010000)_{2s}$
b. $(011010)_{2s} - (001101)_{2s}$
c. $(000011)_{2s} - (010000)_{2s}$

Solusi:

(a) $(001010)_{2s}$ (b) $(001101)_{2s}$ (c) $(110011)_{2s}$

2-18. Assuming a 6-bit 1's complement system, perform the following subtraction operations by converting it into addition operations:

a. $(011111)_{1s} - (010101)_{1s}$
b. $(001010)_{1s} - (101101)_{1s}$
c. $(100000)_{1s} - (010011)_{1s}$

Solusi:

(a) $(001010)_{1s}$ (b) $(011100)_{1s}$ (c) Overflow

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Using 8-bit signed magnitude:

a) $0100\ 1111_2 + 0010\ 0011_2$

b) $0100\ 1111_2 + 0110\ 0011_2$

c) $(99)_{10} - (79)_{10}$

d) $(-19)_{10} + (13)_{10}$

e) Subtract $(-24)_{10}$ from $(-43)_{10}$

Solusi:

Delapan kondisi operasi ADD/SUBRACT pada sistem bilangan SIGNED-MAGNITUDE

Operation	ADD Magnitudes	SUBTRACT Magnitudes		
		A > B	A < B	A = B
(+A) + (+B)	+ (A + B)			
(+A) + (-B)		+ (A – B)	- (B – A)	+ (A – B)
(-A) + (+B)		- (A – B)	+ (B – A)	+ (A – B)
(-A) + (-B)	- (A + B)			
(+A) - (+B)		+ (A – B)	- (B – A)	+ (A – B)
(+A) - (-B)	+ (A + B)			
(-A) - (+B)	- (A + B)			
(-A) - (-B)		- (A – B)	+ (B – A)	+ (A – B)

A dan B adalah magnitude (7 bit) tidak termasuk signed bit (1 bit) penanda +/-

(a) Operation: (+A)+(+B) → $(01110010)_2$

(b) Overflow (Out of range) → $(01001111)_2$+$(01100011)_2$ = $(010110010)_2$ (bit signed-magnitudenya berlebih menjadi 9 bit, melebihi 8-bit signed magnitude system).

Jika perhitungan dilakukan dalam desimal → $(79)_{10}$ + $(99)_{10}$= $(178)_{10}$ → melebih range 8 bit signed magnitude: -127 to +127

(c) Operation: (+A)-(+B) ; A > B ; +(A-B) → $(1100011)_2$-$(1001111)_2$=$(0010100)_2$ tambahkan signed bit positif → $(00010100)_2$

(d) Operation: (-A)+(+B) ; A > B ; -(A-B)→ ubah $(-19)_{10}$ ke $(19)_{10}$ →-($(19)_{10}$ - $(13)_{10}$) → $(0010011)_2$ -$(0001101)_2$ =$(0000110)_2$ tambahkan signed bit negatif → $(10000110)_2$

(e) $(-43)_{10}$ – $(-24)_{10}$ → Operation: (-A)-(-B) ; A > B ; -(A-B)→-($(43)_{10}$-$(24)_{10}$)= $(0101011)_2$ – $(0011000)_2$ = $(0010011)_2$ tambahkan signed bit negatif → $(10010011)_2$